

Chen-Rui Xia

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EDUCATION

Peking University

PhD candidate in Center for Bioinformatics;

Advisor: Dr. Ge Gao

Beijing, China

Sep 2021 – Present

Harbin Institute of Technology

Bachelor of Bioinformatics;

Harbin, China

Sep 2017 – June 2021

PUBLICATION

Google Scholar: <https://scholar.google.com/citations?user=2scMf9oAAAAJ&hl=en>

- **Chen-Rui Xia***, Jia-Wen Yao*, Ge Gao[†]. Quantifying predictability of gene expression from histology image, **2025. (In submission)**
- **Chen-Rui Xia***, Zhi-Jie Cao[†], Ge Gao[†]. DECIPHER for learning disentangled cellular embeddings in large-scale heterogeneous spatial omics data, *Nature Communication*, **2025 (Editors' Highlight)**. ([Github](#))
- **Chen-Rui Xia***, Jun-Yan Qian*, Yang Ding*, Jiu-Liang Zhao, Qian Wang, Xin-Ping Tian, Xiao-Feng Zeng, Ge Gao[†], Meng-Tao Li[†]. Global immune repertoire profiling suggests multifaceted unresolved dysregulation upon systemic lupus erythematosus immunosuppressant therapy, **2025. (Under review)** ([Github](#))
- Chen-Kai Guo*, **Chen-Rui Xia***, Guang-Dun Peng, Zhi-Jie Cao[†], Ge Gao[†]. Learning phenotype associated signature in spatial transcriptomics with PASSAGE, *Small Methods*, **2025**. ([Github](#))
- **Chen-Rui Xia***, Zhi-Jie Cao*, Xin-Ming Tu, Ge Gao[†]. Spatial-linked alignment tool (SLAT) for aligning heterogeneous slices, *Nature Communication*, **2023 (Editors' Highlight)**. ([Github](#))
- Xin-Ming Tu*, Zhi-Jie Cao*, **Chen-Rui Xia**, Sara Mostafavi[†], Ge Gao[†]. Cross-Linked Unified Embedding for cross-modality representation learning, *NeurIPS*, **2022 (Oral Presentation, top 5%)**. ([Github](#))

AWARDS

- Rising Star of General Artificial Intelligence, Peking University 2025
- National Center for Advanced Training in Biology, Peking University 2025
- Peking University President Scholarship 2025&2024
- China National Natural Science Foundation for Ph.D. students (*first batch*) 2024
- Novogene Scholarship 2024
- Graduate Research Star of School of Life Science 2023
- Outstanding Graduates of Harbin Institute of Technology 2021

ACADEMIC SERVICE

- Reviewer of *Nature Methods*, *Nature Computational Science*, *Nature communication* and *Genome Biology*.

SKILLS

Coding: Python, Slurm, PyTorch

Other: Badminton